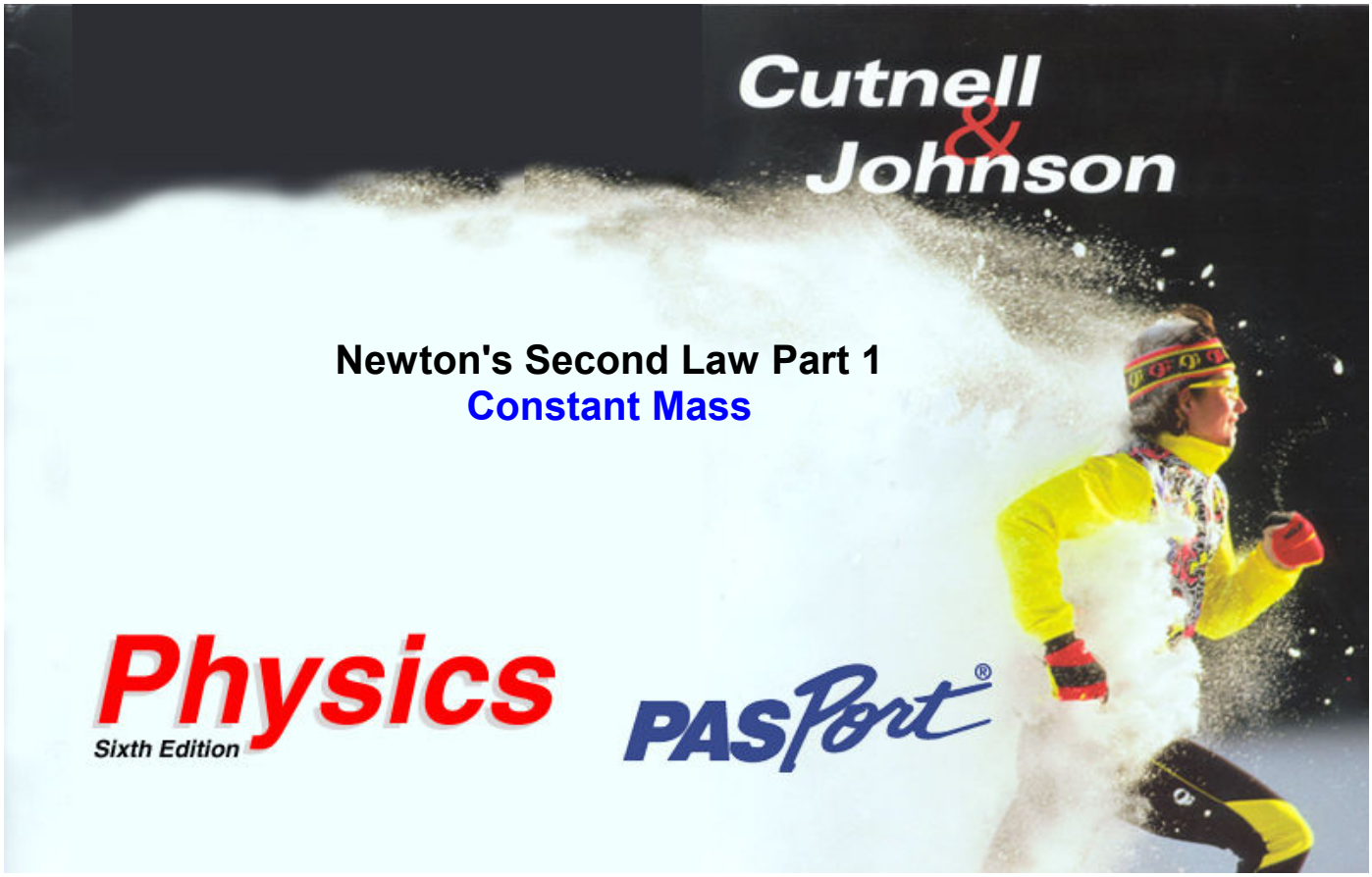

Cutnell & Johnson
Newton's Second Law Part 1
Constant Mass
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

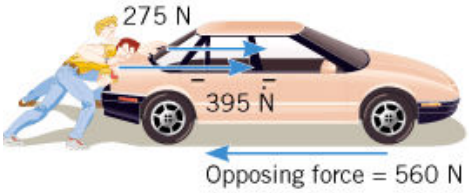
Date:	
Your Name:	
Group:	
Period:	

Introduction *Exploration Goals*

§ One of the important equations in physics is a = F/m, also known as Newton's Second Law of Motion. Newton's Second Law describes the behavior of everything that changes its motion due to a net force -- from the trajectory of a baseball to the motion of a planet.

§ The purpose of this exploration is to find out what happens to an object's acceleration when the net force applied to the object increases and the mass of the system is constant.

§ Use a Motion Sensor to measure the motion of an object that is accelerated by a net force. Determine what happens to the acceleration of the cart when the net force is increased and the mass stays constant.



Note: In Newton's Second Law Part 2 you will measure the acceleration of an object when the net force is kept constant but the mass is increased.

Learning Outcomes

You will be able to:

- Interpret data from a graph of velocity versus time to find acceleration.
- Measure the acceleration of an object with a constant mass when the net force is changed.
- Calculate theoretical values for acceleration based on net force and mass and compare the theoretical values with the measured values.
- Describe the relationship between the acceleration of an object and the net force applied to the object.

Hypothesis

§ What happens to an object when you apply a net force to it?

§ What happens to the motion when you change the magnitude of that net force?

Background

§ Newton's first law states that if no net force acts on an object, then the velocity of the object remains unchanged. The second law deals with what happens when a net force does act.

§ As long as a net force acts, the velocity of an object changes - in other words, it accelerates. If more force is applied, the greater force produces a greater acceleration. Twice the force produces twice the acceleration.

§ Often, several forces act on an object simultaneously. In such cases, it is the net force, or the vector sum of all the forces acting, that is important. Newton's second law states that the acceleration is proportional to the net force acting on the object.

§ Newton's Second Law also states that the acceleration is inversely proportional to the mass.

$$a \approx F_{net}$$

» Acceleration is directly proportional to net force.

$$a \approx \frac{1}{m}$$

» Acceleration is inversely proportional to mass.

$$a = \frac{F_{net}}{m}$$

» Newton's Second Law of Motion

Materials

- PASPORT USB Link
- PASPORT Motion Sensor
- Mass and Hanger Set
- PAScar
- Super Pulley with Table Clamp
- String (about 1.2 m)
- Balance
- 1.2 m Dynamics Track

Setup *Equipment Setup*

Safety Reminder: Follow the directions for using the equipment.

» You can use the same setup for both Part 1 and Part 2 of Newton's Second Law.

§ The Dynamics Track has a fixed end stop at one end. Place the Motion Sensor on the track in front of the fixed end stop.

§ Attach the Pulley to the other end of the Dynamics Track. If you have a piece of wood you can place this in front of the pulley to stop the cart.

§ Level the track. Put the cart on the track. If the cart moves one way or the other, use the leveling screw under the fixed end stop to level the track.

§ Attach a piece of string about 1.2 m long to one end of the cart.

Note: It is very important that the track is level to get the best results.

Setup *Equipment Setup*

§ Add a 20-g mass to the mass hanger and weigh the hanger plus the mass. Record the mass.

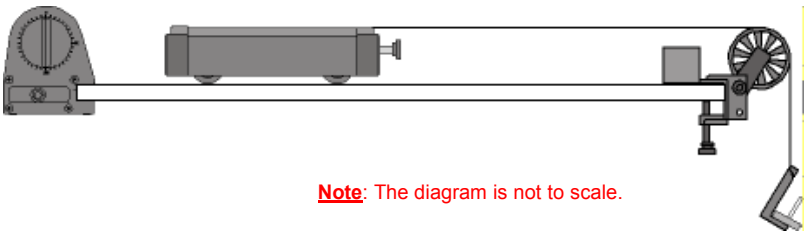
§ Put two 20-g masses into the accessory tray on top of the cart.

§ Weigh the cart plus the two 20-g masses and record the total mass of the cart and masses.

§ Attach the mass hanger to the other end of the string, and put the string in the pulley's groove.

» Run #1: Total mass of the hanger and 20-g mass (mH):

» Run #1: Total mass of the cart and two 20-g masses (mC):



Note: The diagram is not to scale.

Process *Prepare to Record Data: Constant Mass*

Read this page before you take data. The next page is where you will record data.

§ You will take three runs of data. For Run #1, use the arrangement described on the previous page (one 20-g mass on the mass hanger and two 20-g masses on top of the cart).

» For the second and third runs, transfer mass from the tray on the cart to the mass hanger on the string.

§ For Run #2, transfer one 20-g mass from the tray of the cart to the mass hanger before you record data. The total mass of the system is constant, but the net force is increased.

§ For Run #3, transfer a second 20-g mass from the tray of the cart to the mass hanger before you record data. As in Run #2, the total mass of the system is constant, but the net force is increased.

Note: Catch the cart before it hits the pulley or you could damage the pulley.

Process *Record Data: Constant Mass, Increasing Net Force*

§ Click **Start**. Release the cart. Stop the cart before it hits the pulley and click **Stop**.

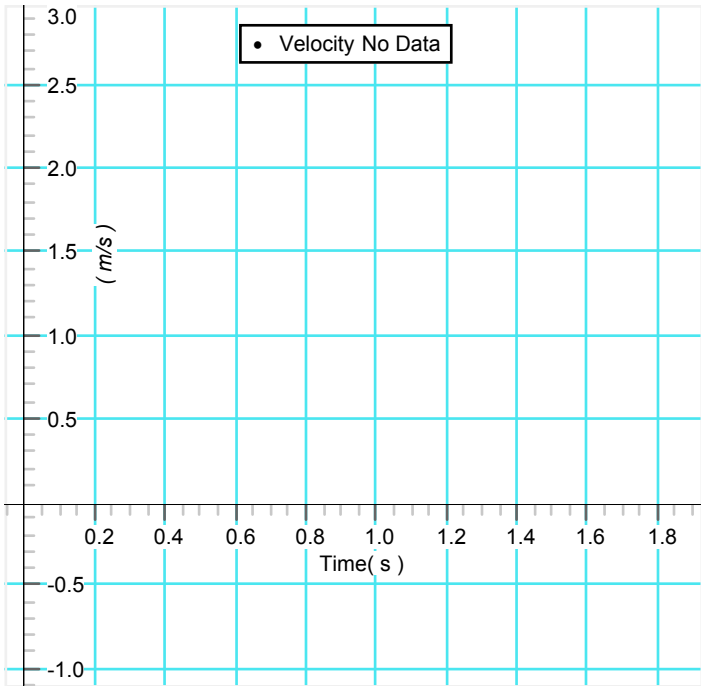
» The next page describes how to determine the acceleration.

§ For Run #2, move a 20-g mass from the cart to the hanger. Record the total mass of the hanger. Record data as before.

§ For Run #3, move the remaining 20-g mass from the cart to the hanger. Record the total mass of the hanger.

» Run #2: Total of hanger plus masses

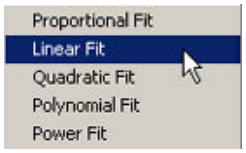
» Run #3: Total of hanger plus masses



Analyze *How to Find the Acceleration from the Velocity*

§ Click **Scale to Fit** to rescale the graph if needed. Use the cursor to highlight a smooth part of the graph.

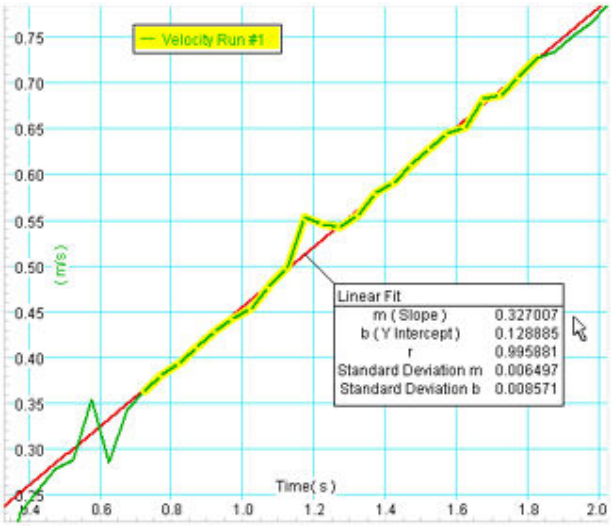
§ Click the **Fit** menu and choose **Linear Fit**.



§ The slope "m" of the best fit line is the acceleration.

§ After recording the slope, use the **Data** menu to select the next run.

» Analyze your data on the next page.



This example shows the highlighted area and the Linear Fit information box. Note that the Scale to Fit button, Fit menu, and Data menu are not shown.

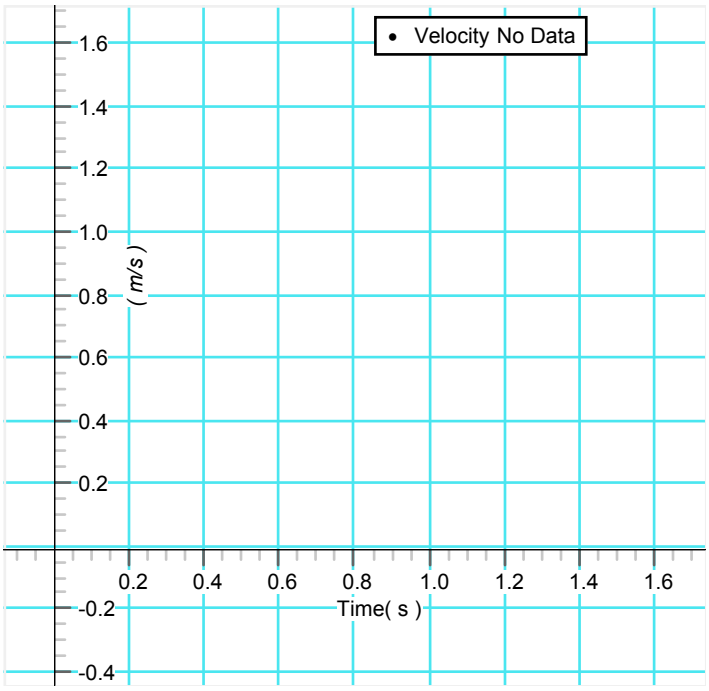
Analyze *Determine the Acceleration from the Velocity*

§ Select **Run #1**. Click **Scale to Fit** if needed. Highlight part of the graph. Select **Linear Fit** from the **Fit** menu.

§ Record the value of "m" (slope) as the acceleration for Run #1.

§ After recording the slope, use the **Data** menu to select the next run.

§ Repeat the process for Run #2 and Run #3



» The slope "m" of the best fit line is the acceleration.

Run #1 acceleration

Run #2 acceleration

Run #3 acceleration

Observations

§ What did you observe about the slope of the Linear Fit as the net force increased but the total mass was kept constant?

§ Why did the slope change for each run?

§ What variable(s) did you keep the same from one run to the next? What variable(s) did you control?

§ What variable changed as a result and what was the response?

Analyze *Calculations: Constant Mass, Increasing Net Force*

§ Calculate the theoretical acceleration when mass is constant and net force is changed.

§ The acceleration is the ratio of the net force divided by the total mass. For the first three runs, the total mass of the system (mass of cart plus mass of hanger) is constant and the net force (mass of hanger x 9.8) increases.

§ Assuming no friction, the net force is the weight of the hanger (mass x 9.8).

$$a = \frac{m_H g}{m_c + m_H}$$

g = 9.8 m/s/s

	Total mass of the hanger (mH)	Net force (mass of hanger x 9.8)(mHg)	Acceleration (theoretical)
Run #1:			
Run #2:			
Run #3:			

Total mass of cart and two 20-g masses(mC):

Analyze *Comparison*

§ Compare the measured acceleration to the theoretical acceleration for each run. Find the percent difference (% Difference).

	Measured	Theoretical	% Difference
Run #1:			
Run #2:			
Run #3:			

$$\% \text{ difference} = \left| \frac{\text{measured} - \text{theoretical}}{\text{theoretical}} \right| \times 100$$

§ How do your results compare to your hypothesis?

Hypothesis

Synthesize *Conclusions*

§ What happens to an object's acceleration if the net force applied to the object is increased but the object's mass remains constant?

§ Is the object's acceleration proportional to the net force?

§ What were the sources of error in this experiment?

Synthesize *Applications and Extension Problem*

§ Based on Newton's Second Law, predict what will happen to the Space Shuttle's acceleration after lift off as it burns its fuel.

§ Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 4, Problem 3, page 116.

In the amusement park ride known as Magic Mountain Superman, powerful magnets accelerate a car and its riders from rest to 45 m/s (about 100 mi/h) in a time of 7.0 s. The mass of the car and riders is 5,500 kg. Find the average net force exerted on the car and riders by the magnets.